Cambodia rock and geochemistry atlas

Maps, authentic photographs, classification references and extracted analytical tables

This atlas supplements your rock-classification document with inspectable evidence. It combines your geological and structural GIS with official JICA analytical tables, CCOP/MME specimen and field photographs, and published Snoul outcrop, core and microscope images.

What is included

42 geological descriptions retained from your supplied national map; 74 located JICA rock samples; 20 whole-rock major-element analyses; 51 rock analyses and 102 sediment analyses for 36 elements. The package contains the original sources, extracted tables, figures and a download manifest.

Coverage is documented, not assumed

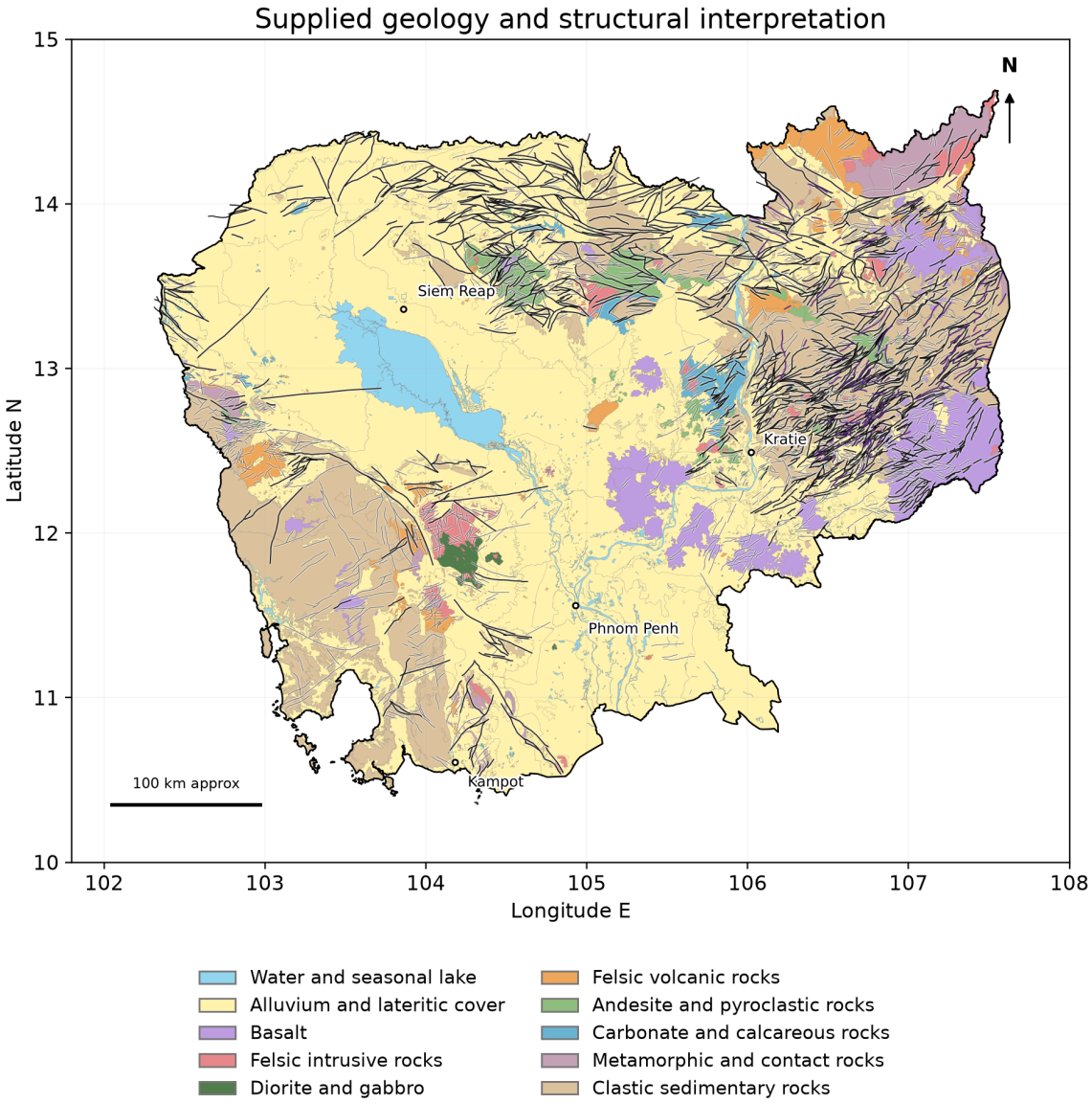
Verified Cambodian photographs are included where available. All 19 photograph requests from your document are tracked individually. Several exact rock/locality combinations remain unfilled. A mapped rock name does not verify a photograph, and a weathering profile is not a fresh hand specimen.

Classification rule

QAPF requires measured modal mineral proportions. JICA whole-rock oxide chemistry cannot supply those proportions directly. The QAPF chart is provided as an authoritative reference without invented Cambodian sample points. Chemical plots show actual reported measurements and their limitations.

Original attachment and GIS retained unchanged. Source IDs, sample IDs and licenses accompany the evidence.

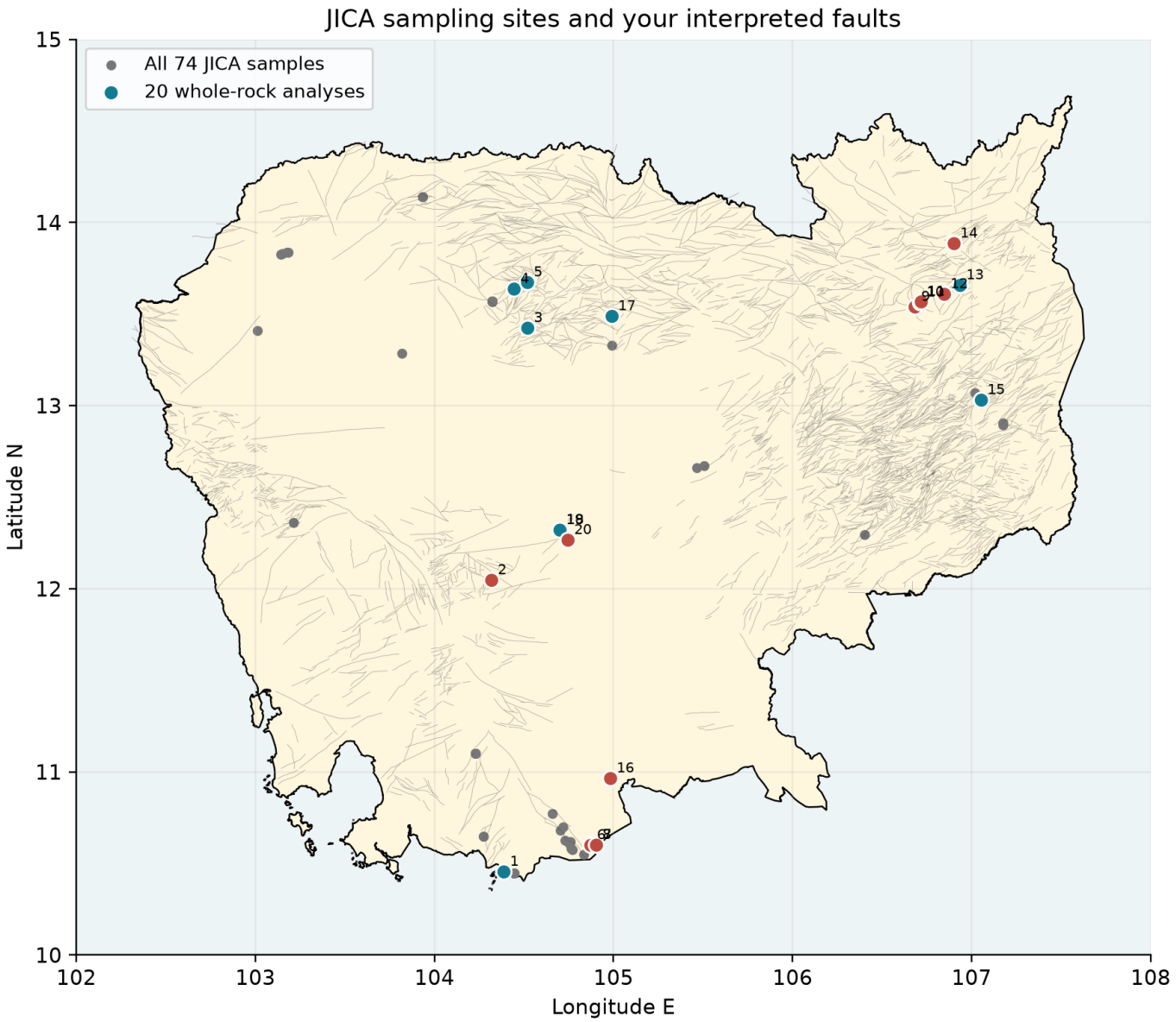
Your geological and structural map



Your national geological polygons and interpreted structures. Conventional lithological colors distinguish intrusive, volcanic, carbonate, clastic, metamorphic and surficial groups; these colors do not encode geophysical values.

User GIS: D:/1-GIS Data/Dept_Geology/1to500kgeomap/geology500k.shp; detailed paths/hashes: GIS_sources.json

Where JICA collected the rock samples



Grey points: all 74 samples. Colored points: the 20 whole-rock analyses; numbers match the analytical index. Red: source-named plutonic rocks; teal: volcanic/dyke/porphyrite samples. Dates and ages are not inferred from colors.

JICA 2010 | Final report Appendix V | 12006003_05.pdf | V-5, PDF pages 131-133 | Datum not stated in source table

What the mapped rocks mean

Igneous rocks

Your GIS contains granite, granodiorite, diorite/gabbro, rhyolite, dacite, andesite, basalt and pyroclastic descriptions. These distinguish intrusive textures from volcanic and fragmental textures. JICA confirms sampled felsic and mafic rocks, but the chemistry collection is small and includes altered material; it is not a national average.

Sedimentary rocks and surficial material

Sandstone, shale, conglomerate, limestone and marl record different depositional materials. Alluvium and lateritic cover are mapped separately from consolidated bedrock. The CCOP catalogue verifies Cambodian carbonate specimens and a calcareous-sandstone outcrop; it does not establish that these are Angkor building-stone sources.

Metamorphic and contact rocks

Your map includes amphibolite/quartzite/cipolin/gneiss, hornfels, skarn and metaconglomerate descriptions. JICA lists hornfels, skarnized sandstone, granitic cataclasite and a sample described as basic hornfels or green schist. These observations do not establish the age or extent of a national crystalline basement.

Structures and mineralization

Your interpreted faults and fold axes provide a spatial framework. JICA photographs document fissure-controlled alteration, quartz-limonite veinlets and a granite contact with hornfels/skarn. Such evidence supports local structural or contact control. Neither a mapped lineament nor a single anomalous assay establishes an economic deposit.

Evidence: user GIS attributes; JICA V-4 and V-5; CCOP catalogue records. Interpretations are deliberately local.

Carbonate rock specimens

Verified Cambodia examples from Banteay Meanchey, not Kampot

Shaly limestone / black shale

CAM_GDP_0050_BMC1901a. Catalogue: Permian; grey to dark grey, hard and fractured. The description is retained as a source identification, not a new petrographic determination.

Limestone

CAM_GDP_0051_BMC1901b. Catalogue: Permian; grey to light grey, hard and fractured. Scale cards remain in the original photographs.

Curation: Department of Petroleum Exploration, General Department of Petroleum, Ministry of Mines and Energy. Registrant: SOR Sokun, 2 September 2024. These photographs fill Cambodian carbonate examples, while the exact Kampot specimen slot remains open.

CCOP GeoSample IDs 000520 and 000519 | Copyright CCOP | CC BY-NC

Full record and download URLs: source_downloads.json

Calcareous sandstone in the field

CAM_GDP_0003, Phnom Dey, Banteay Meanchey. The catalogue describes medium-grained calcareous sandstone interbedded with limestone and coaly siltstone, Carboniferous-Permian. This is an outcrop photograph, not an Angkor sandstone hand specimen. Native source image is small (about 28 KB).

CCOP GeoSample 000377 | Registrant Chhun Chandara, 28 August 2024 | CC BY-NC; copyright CCOP

Weathered basalt and lateritic red soil

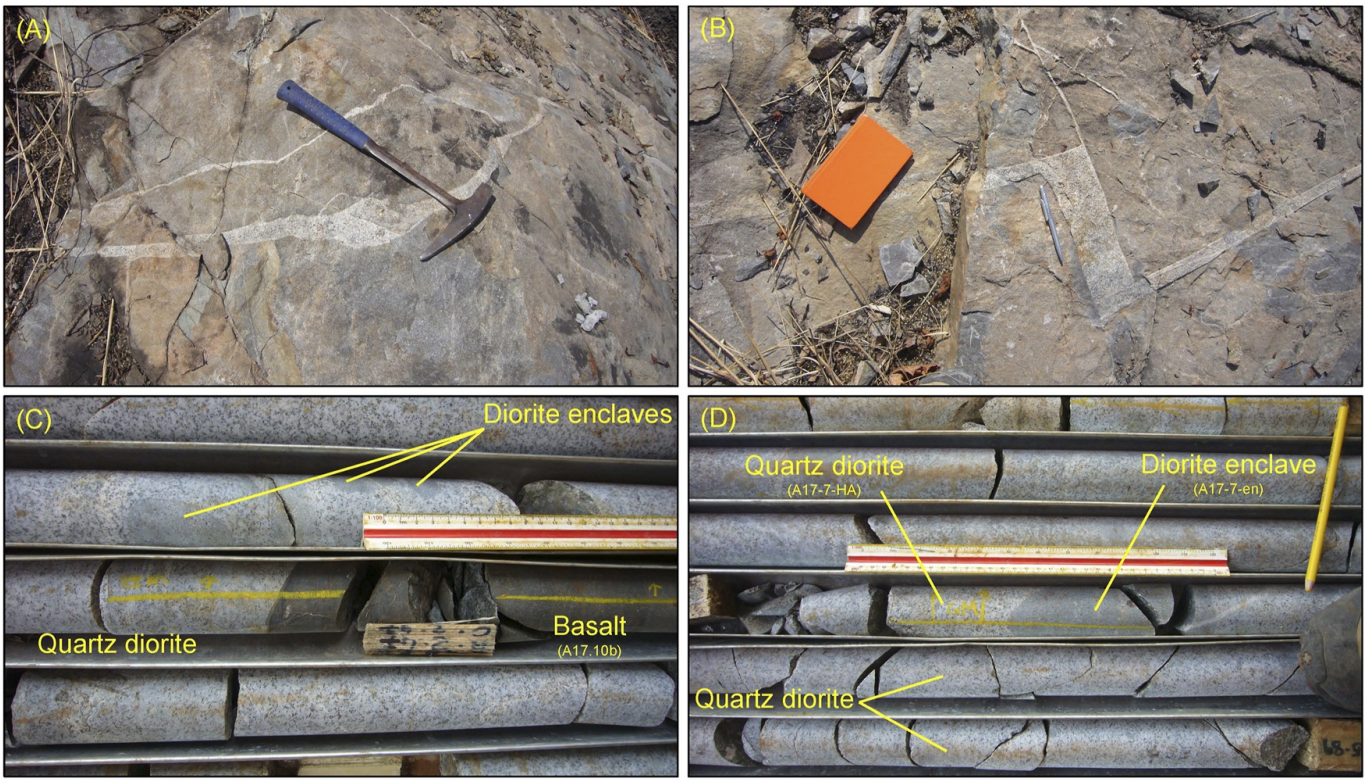
Dak Dam commune, O Raeng district, Mondulkiri

Top: MDK25015, catalogued as soil over weathered basalt. Bottom: MDK25016, lateritic red-soil profile. Neither photograph proves fresh basalt composition, a Ratanakiri locality, or bauxite grade.

CCOP GeoSample 2026-000952 and 2026-000951 | CC BY-NC; copyright CCOP

Original photographs and catalogue HTML retained. Metadata dates differ from filename dates; neither was silently corrected.

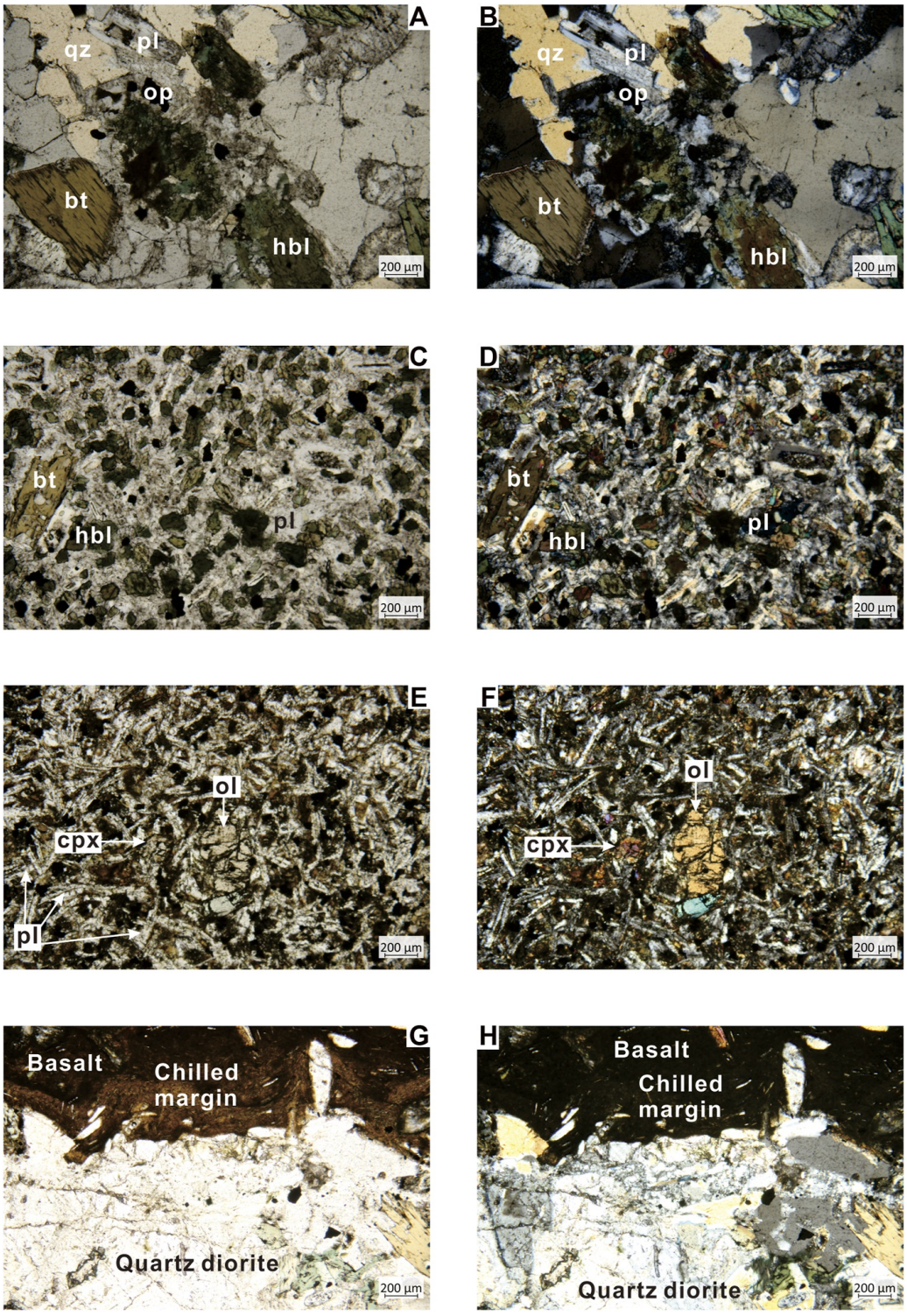
Snoul diorite outcrops and drill core



Published Figure 3. A-B: diorite outcrops and dykes. C-D: diorite enclaves and basalt within quartz-diorite drill core. Original panel labels and scale references remain intact. These are verified Snoul examples, not generic textbook photographs.

Shellnutt et al. 2023 | doi:10.3389/feart.2023.1110568 | CC BY

Snoul petrographic evidence



Published Figure 4: thin-section photomicrographs of the studied rock assemblage. These images help check textures and mineral relationships. They do not supply a complete measured Q-A-P mode for every sample.

Shellnutt et al. 2023 | doi:10.3389/feart.2023.1110568 | CC BY

JICA field photographs 1 to 6

Appendix V-4 Photos of Geological Survey

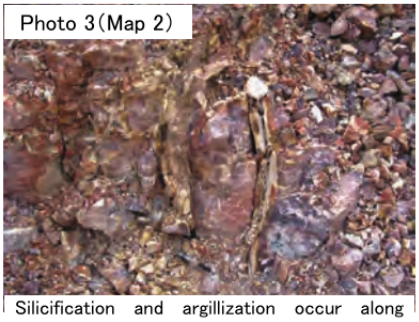
Photo 1 (Map 1)

Silicified andesite or shale with sporadic metallic minerals occurs on this hill.

Photo 2 (Map 2)

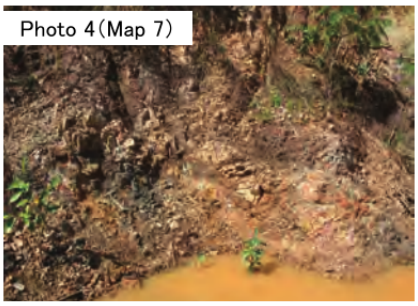
White altered rock with limonite gossan.. The site is a working quarry.

Photo 3 (Map 2)



Silicification and argillization occur along fissures with limonite gossan.

Photo 4 (Map 7)



Limonite gossan and white argillization occur in a pit beside a road.

Photo 5 (Map 9)

Quartz-limonite parallel veinlets occur in conglomerate. Granites and rhyolites occur nearby.

Photo 6 (Map 10)

Fe-Mn rich vein accompanied by quartz, 0.5m in width x 15m in length, filled a fissure in shale.

Apx V-4 1

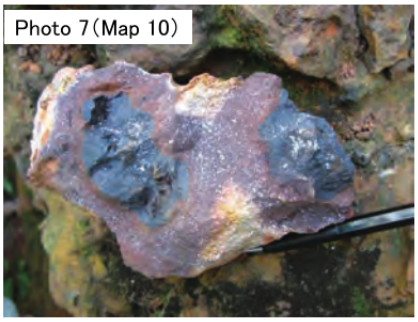
Original source plate: altered rocks, gossan, veinlets in conglomerate, and a Fe-Mn-rich fissure in shale. The source explicitly leaves Photo 1 as andesite OR shale; that uncertainty is retained.

JICA 2010 | Final report Appendix V | 12006003_05.pdf | Appendix V-4 page 1, PDF page 129 | No open image-reuse license verified

JICA field photographs 7 to 12

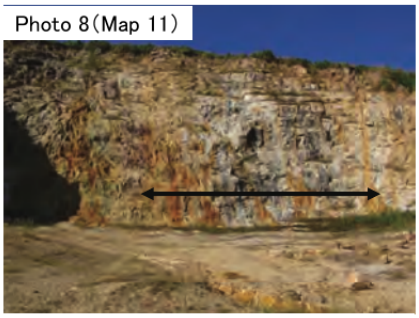
Appendix V-4 Photos of Geological Survey

Photo 7 (Map 10)



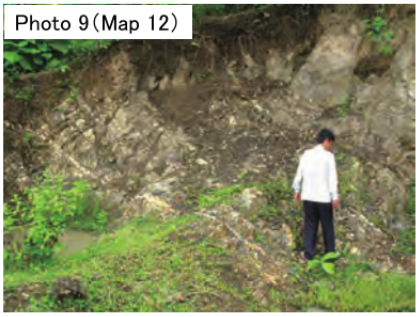
Fe–Mn rich nodules. The black cores are heavier than the brown rims.

Photo 8 (Map 11)



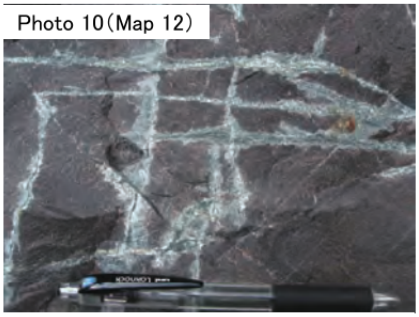
Molybdenite disseminated granite. Black bar shows the disseminated area.

Photo 9 (Map 12)



Hornfels of shale at the contact with granite batholith. Skarnized veinlets with arsenopyrite occur.

Photo 10 (Map 12)

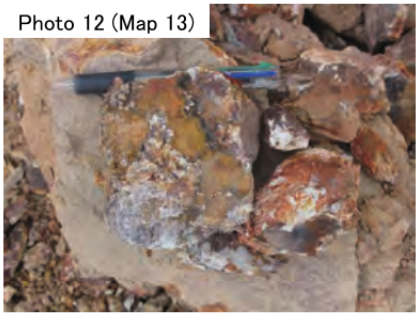


Yellow white skarn nets with amphibole needles and arsenopyrite at the center.

Photo 11 (Map 13)

This hill of quartz porphyry is a gold occurrence. There is a similar gold occurrence 6.5 km away to the east.

Photo 12 (Map 13)



Limonite nets in quartz porphyry.

Apx V-4 2

Original source plate: Fe-Mn nodule; molybdenite-bearing granite; contact hornfels; skarn veinlets; quartz-porphyry hill and specimen. Original photos are low resolution. Map numbers refer to JICA survey maps, not inferred GPS points.

JICA 2010 | Final report Appendix V | 12006003_05.pdf | Appendix V-4 page 2, PDF page 130 | Attributed research excerpts

QAPF classification reference

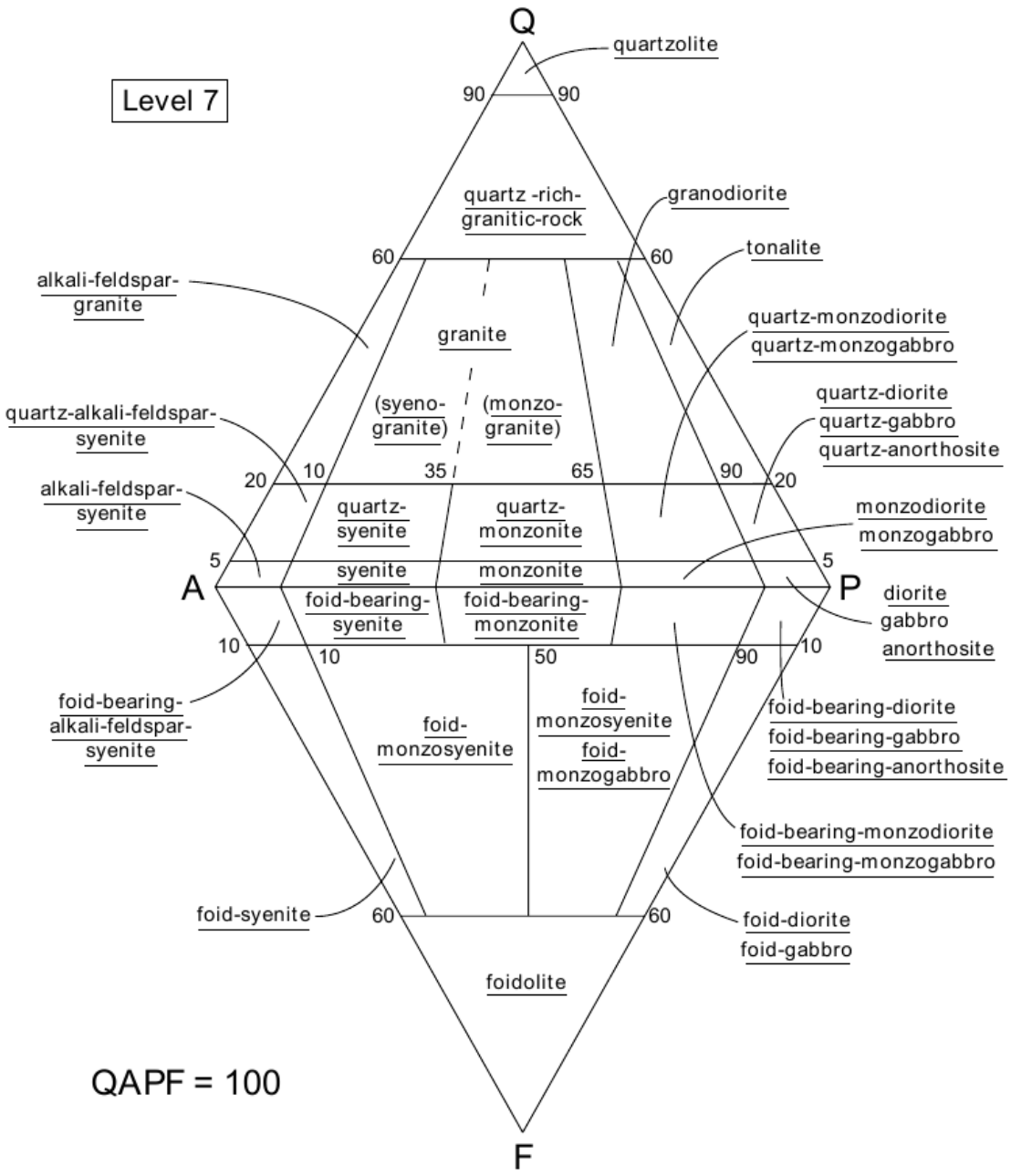


Figure 11 Classification and nomenclature of coarse-grained crystalline rocks according to their modal mineral contents using the QAPF diagram (based on Streckeisen, 1976). The corners of the double triangle are Q = quartz, A = alkali feldspar, P = plagioclase and F = feldspathoid. This diagram must not be used for rocks in which the mafic mineral content, M, is greater than 90%.

BGS Figure 11, cropped with its caption retained. Q = quartz; A = alkali feldspar; P = plagioclase; F = feldspathoids. This is a reference diagram. No Cambodian sample points are plotted because verified complete modal proportions are unavailable.

Gillespie and Styles 1999 | BGS RR/99/006 | figures 11, 12 and 20 | PDF page 37, printed page 35 | Crown copyright

How to use QAPF correctly

Measure the mode first

Use a representative petrographic mode, usually from thin-section point counting or a documented modal measurement. Exclude mafic minerals from QAPF normalization; normalize Q+A+P or A+P+F to 100%. Do not derive a modal quartz or feldspar percentage by reading the bulk SiO2 concentration.

Corrections to the supplied draft

In the BGS/Streckeisen numbered coarse-grained diagram, granite occupies fields 3a and 3b, granodiorite field 4 and tonalite field 5. Diorite/gabbro/anorthosite occupy field 10; quartz-bearing equivalents occupy 10*. The corresponding draft labels should not be reused without correction.

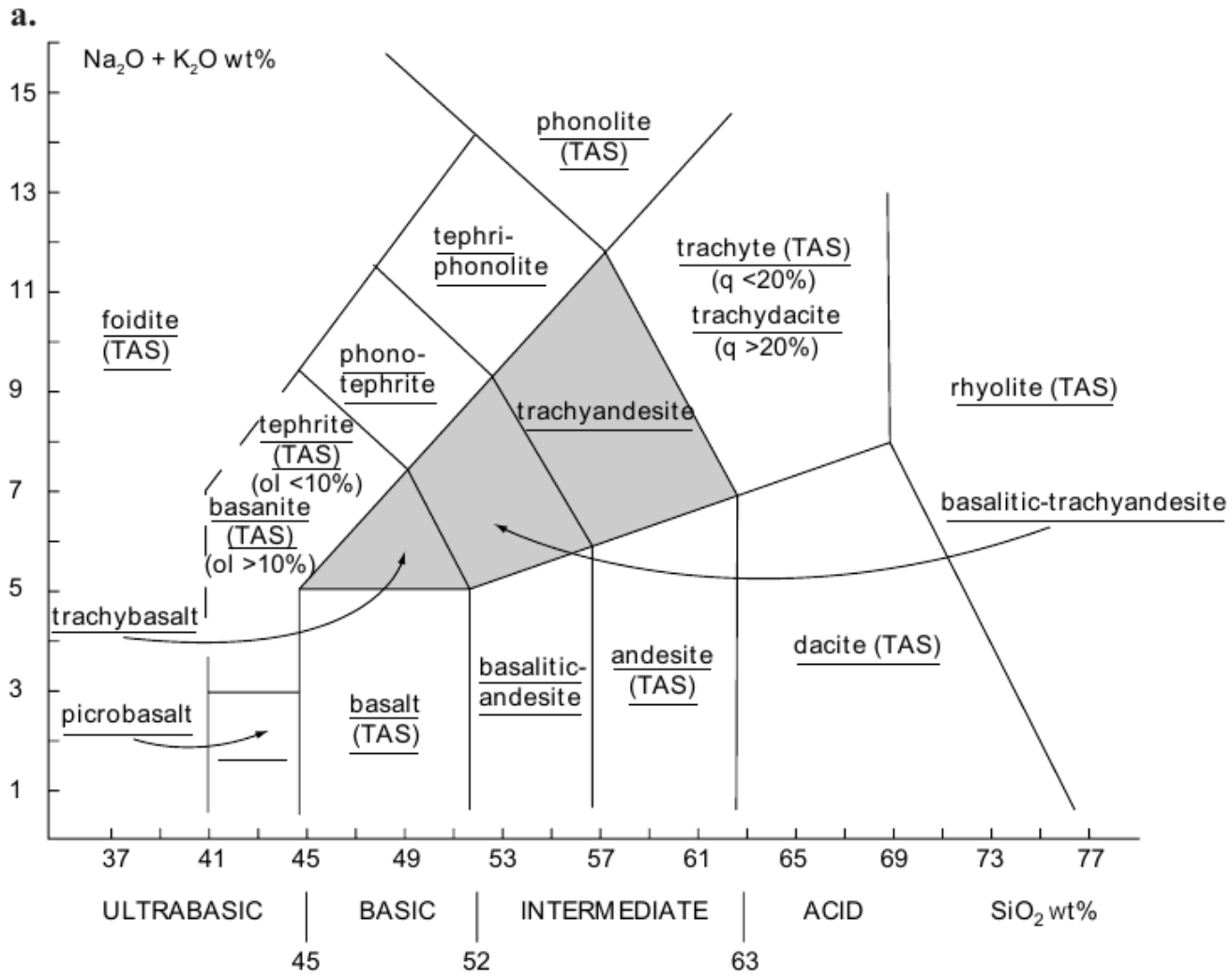
Limits on classification

The chart must not be used for rocks with more than 90% mafic minerals. Diorite and gabbro require additional mineral-composition criteria. Pyroclastic rocks need textural/component classification, and fine-grained or glassy volcanic rocks require a suitable chemical classification when modal data cannot be obtained.

The original BGS report is included for its full classification definitions, numerical boundaries and field numbering. JICA XRD symbols represent qualitative mineral abundance, not measured volumetric percentages, so they were not converted into QAPF sample points.

Gillespie and Styles 1999 | BGS RR/99/006 | figures 11, 12 and 20 | original source PDF included

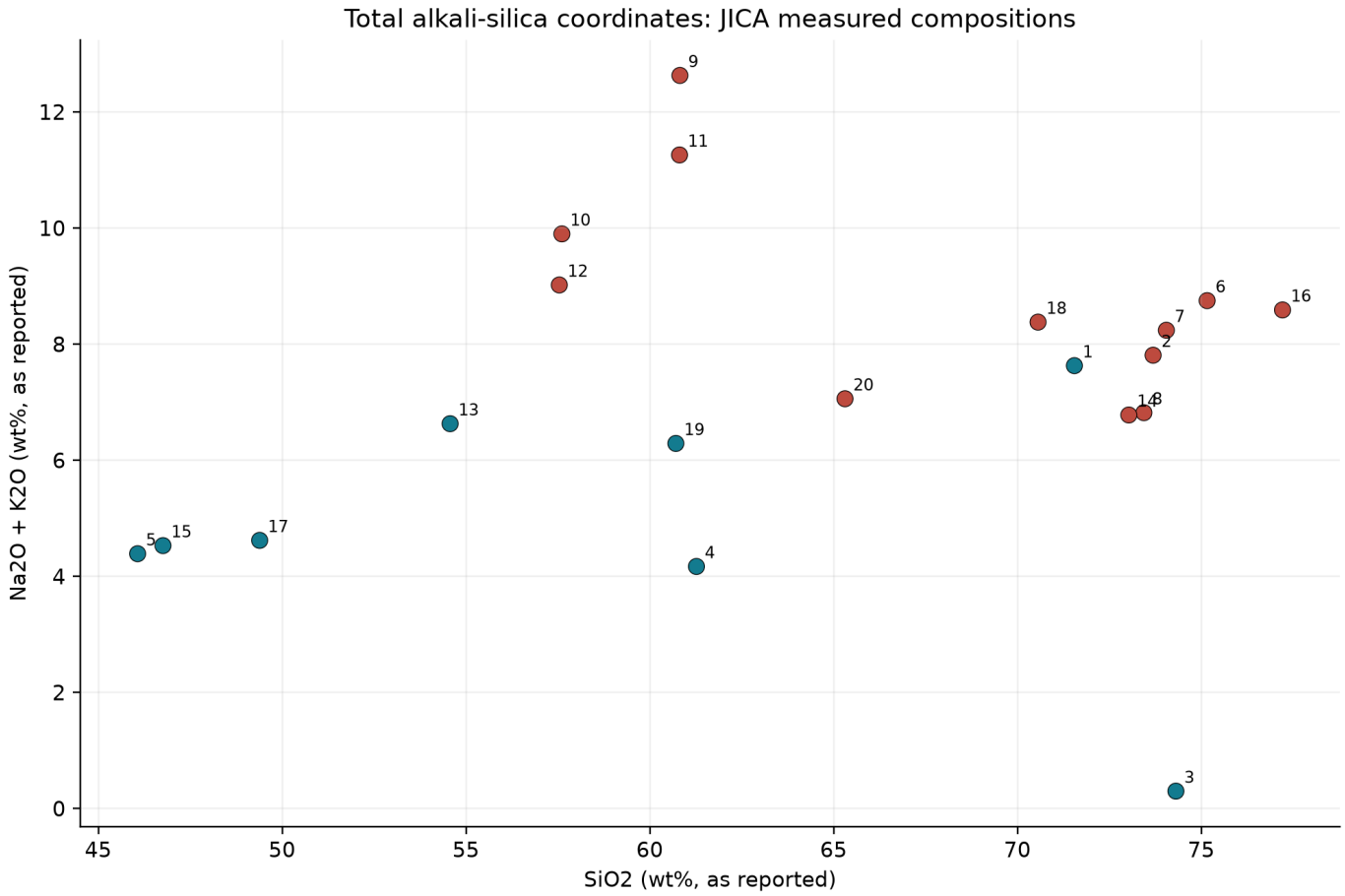
TAS chemical classification reference



BGS Figure 20a, diagram excerpt. TAS is intended for appropriate fine-grained igneous rocks using suitable major-element analyses. Volatile-free normalization and alteration screening are required before assigning formal chemical names. The following measured plot remains separate.

Gillespie and Styles 1999 | BGS RR/99/006 | figures 11, 12 and 20 | PDF page 46, printed page 44 | Crown copyright

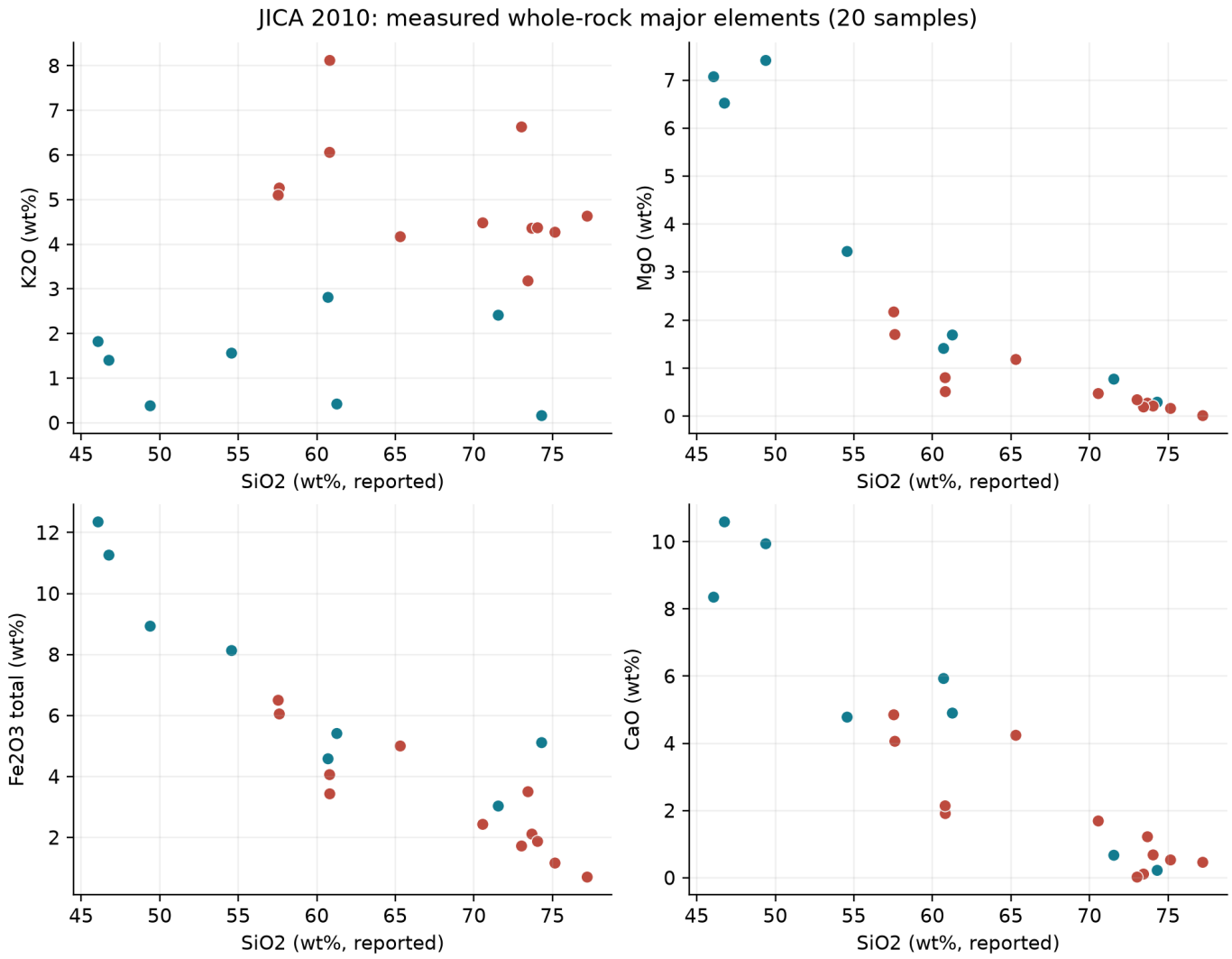
JICA measured alkali and silica contents



Raw reported values, without volatile-free normalization. Every number identifies a sample in the next table. Red denotes source-named plutonic rocks; teal denotes volcanic/dyke/porphyrite samples. Alteration and welded-tuff textures prevent a blanket TAS classification.

JICA 2010 | Final report Appendix V | 12006003_05.pdf | V-7 | 20 samples | SiO2 and Na2O+K2O in wt%

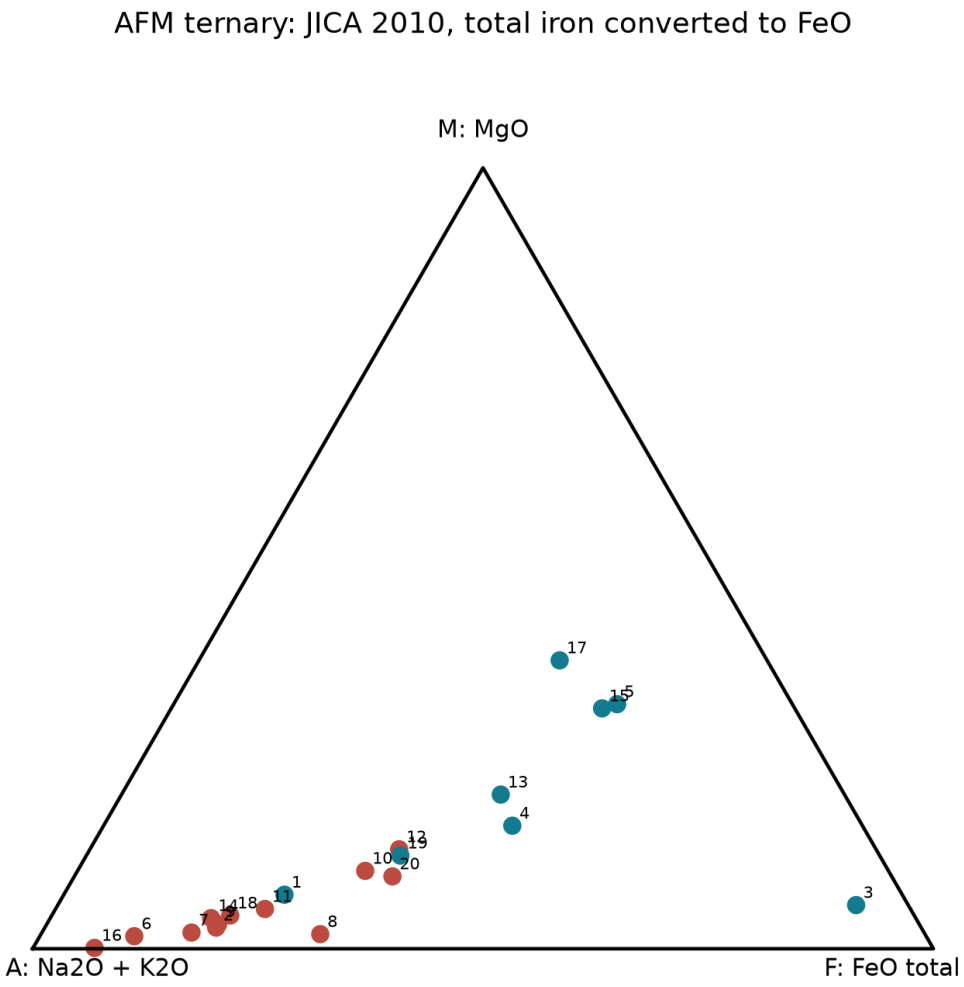
Major element variation



Harker-style plots of actual reported oxides. SiO2 spans 46.07-77.21 wt% in this selected collection. The spread includes distinct rock types and alteration; a common parent magma or differentiation trend is not established by these plots.

JICA 2010 | Final report Appendix V | 12006003_05.pdf | V-7 | All 20 samples shown; no imputation

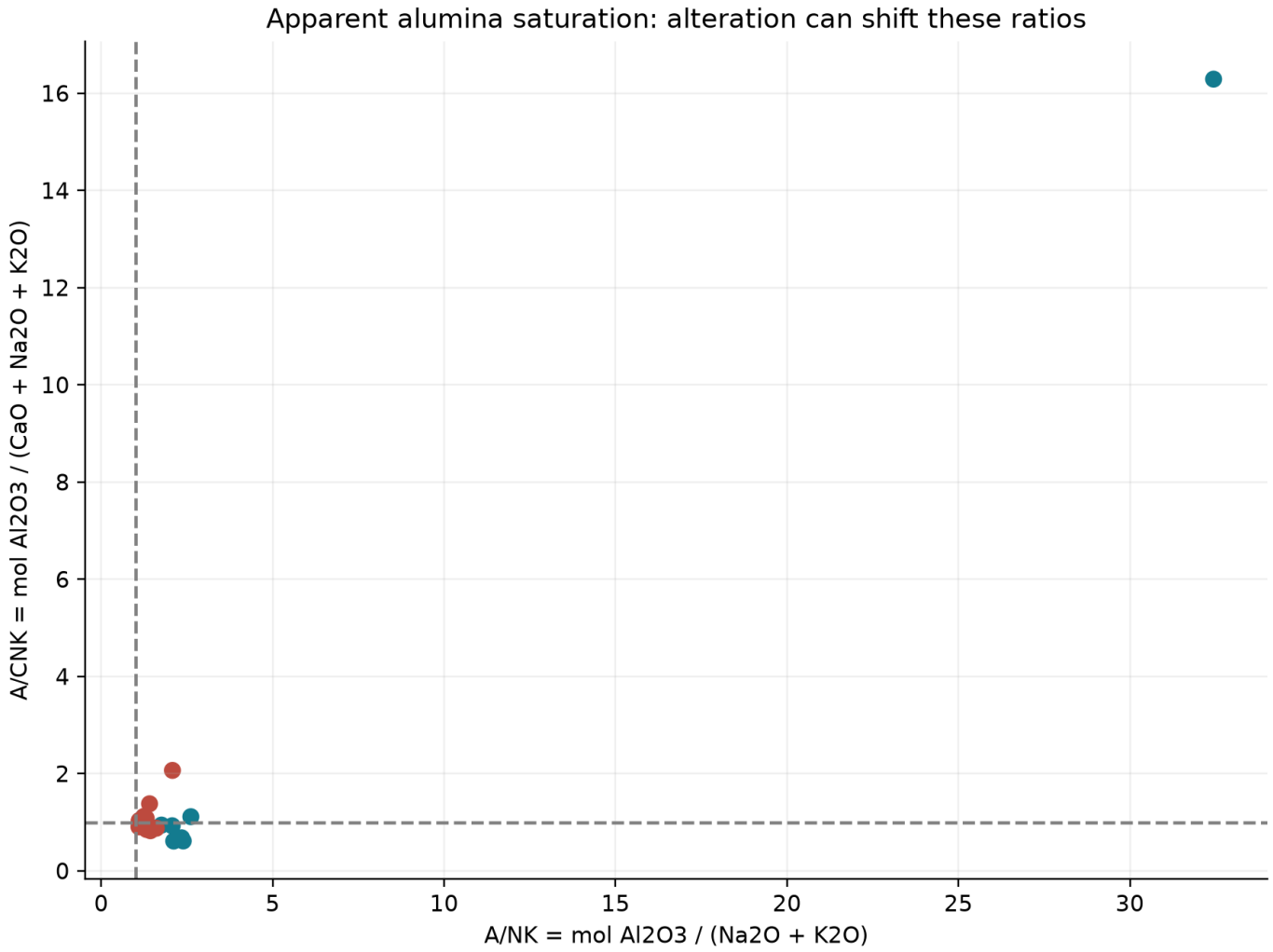
AFM major element proportions



A = Na2O+K2O; F = FeO total, calculated as 0.8998 x reported Fe2O3 total; M = MgO. Components are normalized to their sum for ternary plotting. No tectonic or magma-series boundary has been invented.

JICA 2010 | Final report Appendix V | 12006003_05.pdf | Derived mass proportions; ratios are affected by alteration

Apparent alumina saturation



Molar ratios use Al2O3, CaO, Na2O and K2O. Dashed lines mark unity. These are whole-rock screening ratios without apatite-Ca correction. Alteration can mobilize Na, K and Ca; the plot alone cannot establish I-type/S-type origin or tectonic setting.

JICA 2010 | Final report Appendix V | 12006003_05.pdf | V-7 | Molecular weights and plotted values documented with CSVs

Index of the 20 whole rock analyses

Sample numbers match the map and chemical diagrams

01 9060509 Rhyolite
02 9060802 Granite
03 9061405 Dacite
04 9061407 Augite-hypersthene andesite
05 9061408 Picritic basalt
06 9061702A Granite
07 9061704 Granite or granite porphyry
08 9061705A Mineralized granite
09 9080703 Granite
10 9080704 Granite porphyry
11 9080705 Granite porphyry
12 9080706 Granite (adamellite)
13 9080707 Basalt dyke
14 9080801 Welded tuff
15 9081006 Porphyrite
16 9082302A Pink granite with QV (adamellite)
17 9111301 Basalt
18 9111701 Welded tuff
19 9111702 Andesite
20 9111703 Granodiorite

CSV preserves original oxide values and IDs. Source: JICA 2010 | Final report Appendix V | 12006003_05.pdf | V-7, PDF page 135

Analytical data and quality limits

Major elements

20 samples analysed by X-ray fluorescence. Oxides and loss on ignition are in wt%; total iron is expressed as Fe2O3. LOI is measured at 1000 degrees Celsius. P2O5 is unreported for 9082302A; it remains blank. SrO, BaO and Cr2O3 were reported only for the first eight samples and remain blank elsewhere.

Rock and sediment trace elements

51 rock samples x 36 elements are exported from Appendix V-8. An additional 102 sand/stream-sediment analyses x 36 elements are exported from Appendix V-10. These are different sample media and remain separate. Elemental percentages are not oxide percentages. Censored tokens such as <0.2 remain text; no zero or half-detection-limit substitution was applied.

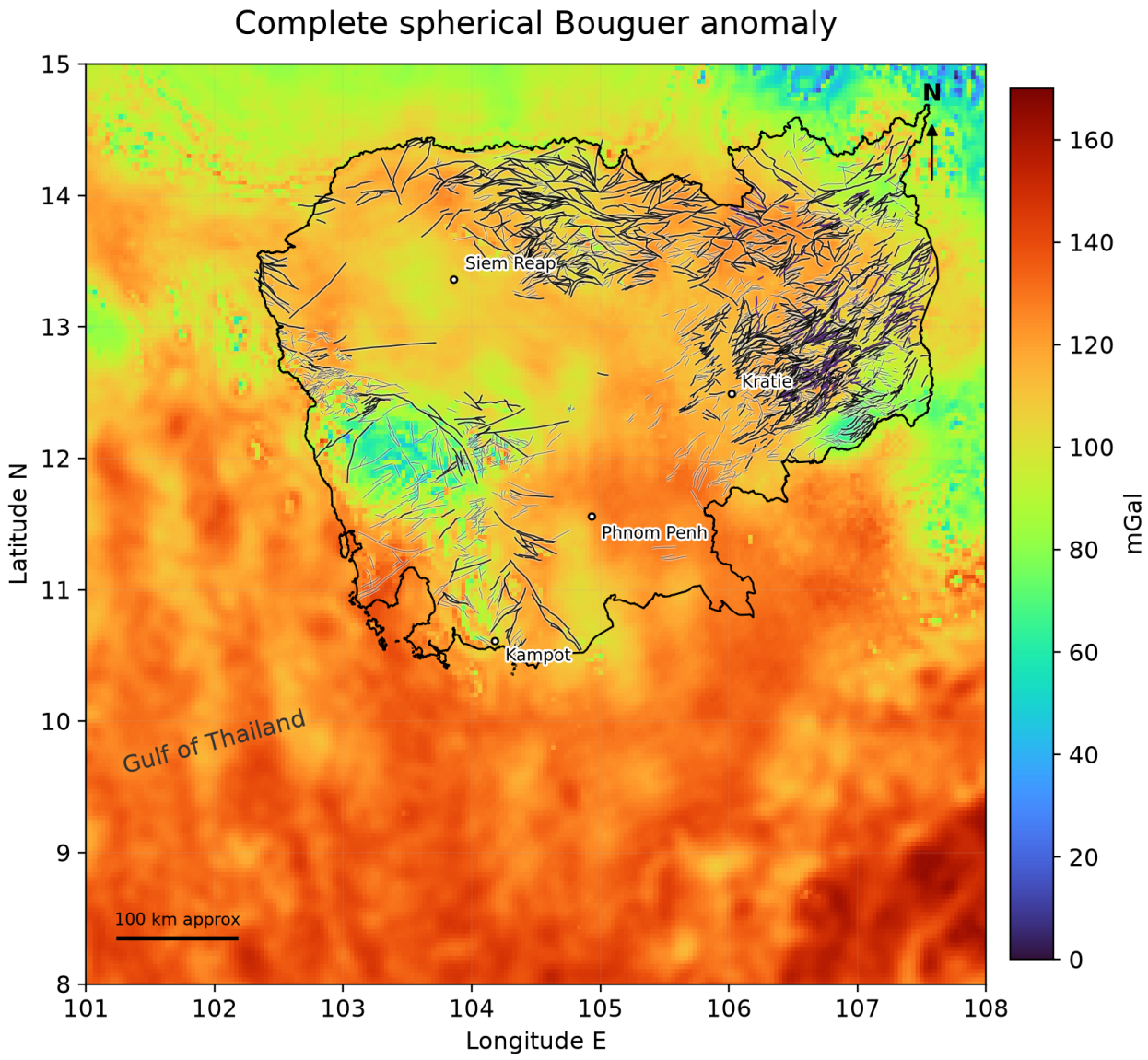
Spatial checks

All 20 major-element samples join the 74-sample inventory by exact sample ID. Inventory coordinates are degrees and decimal minutes converted to decimal degrees; the source does not state a datum. The table appears to repeat Kep-area coordinates for sample 9060909 while listing province Pur. That source discrepancy is retained, and should be checked before detailed spatial interpretation.

The 20 analyses are a selected 2009-era survey collection published in 2010. They are not proof of national abundance or representative provincial composition. Elevated LOI, low alkalis in the source-named dacite, mineralized granite and strongly altered sample descriptions warrant petrographic checking before genetic interpretation.

JICA 2010 | Final report Appendix V | 12006003_05.pdf | V-5 to V-10 | Original PDFs and table text accompany CSVs

Gravity and the geological framework

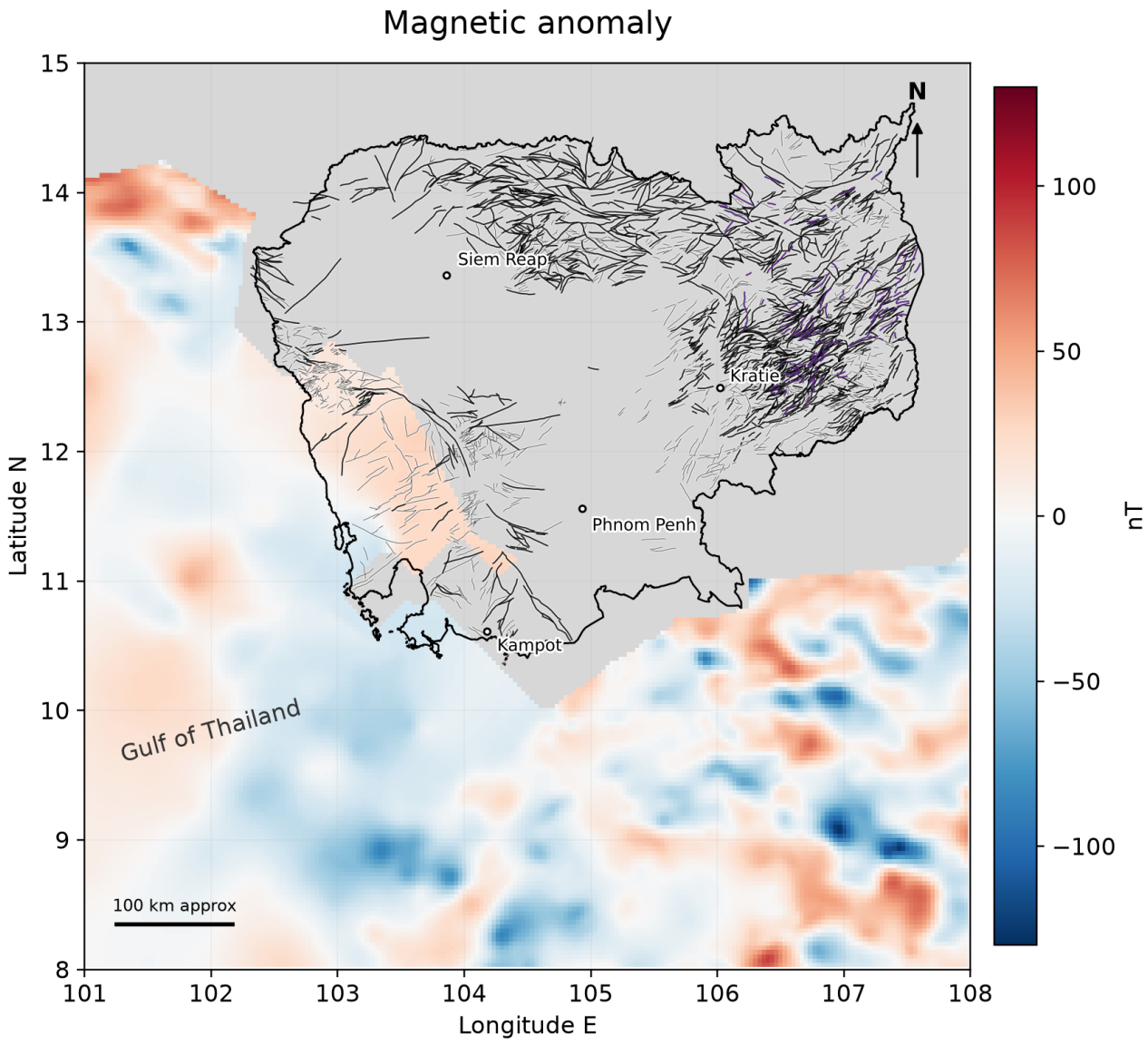


Black lines: supplied interpreted faults | Purple: supplied fold axes
Black outline: supplied administrative boundary | Grey: source NoData

Bouguer gravity provides a regional density framework, shown with your interpreted faults and fold axes. The conventional geophysical rainbow ramp uses cooler colors for lower values and warmer colors for higher values. Gravity does not uniquely identify granite, basalt, a fault, or an orebody.

Existing verified project grid: BGI WGM2012 | mGal | provenance in out/README.txt and prior geophysical report

Magnetic context and its coverage limit



Black lines: supplied interpreted faults | Purple: supplied fold axes
Black outline: supplied administrative boundary | Grey: source NoData

EMAG2v3 magnetic anomaly, red-blue diverging ramp in nT. Grey is source NoData. Only about 10.4% of cells within your supplied Cambodia boundary have valid values in this regional sea-level extract. It cannot support nationwide magnetic lineament or lithology claims.

Existing verified project grid: NOAA EMAG2v3, 2017 | detailed coverage validation in prior geophysical report

Photograph coverage and remaining gaps

Requested slots A1 to A10; exact locality and material matter

A1 Kampot limestone hand specimen | Partial

Cambodian limestone specimen verified from Banteay Meanchey; different locality

A2 Pailin radiolarian chert hand specimen | Gap

No directly downloaded, locality-verified photograph

A3 Angkor feldspathic arenite hand specimen | Gap

Calcareous sandstone outcrop is a different rock and locality

A4 Terrain Rouge arkose or conglomerate hand specimen | Partial

JICA conglomerate outcrop with quartz-limonite veins; not a Terrain Rouge specimen

A5 Laterite or bauxitic laterite | Partial

Lateritic red-soil profile verified; bauxite composition not established

A6 Kampot karst limestone outcrop | Gap - download failed

GEUS report verified online; public PDF timed out twice; no local plate extracted

A7 Phnom Kulen sandstone escarpment | Gap

No directly downloaded locality-verified photograph

A8 Mekong Quaternary alluvium outcrop | Gap

No directly downloaded locality-verified photograph

A9 Phnom Aural or Ba Phnom granite hand specimen | Gap

Uchida 2023 figures located but publisher access blocked

A10 Snoul diorite or quartz diorite specimen | Verified core alternative

Published drill-core specimens, outcrop photographs and micrographs

Complete machine-readable record: data/photo_coverage_19_requested_slots.csv. No synthetic photos or locality substitutions.

Photograph coverage and remaining gaps

Requested slots A11 to A19; exact locality and material matter

A11 Ratanakiri alkaline basalt hand specimen | Gap

Mondulkiri weathered basalt soil is not a fresh Ratanakiri basalt specimen

A12 Zircon or sapphire concentrate from basalt gravel | Gap

Minerals 2018 source located; download rate-limited

A13 Southwest Cambodia granite outcrop | Partial

JICA molybdenite-bearing granite quarry photograph, Map 11; locality source retained

A14 Ratanakiri basalt plateau outcrop | Partial

Dak Dam Mondulkiri basalt-derived weathering profile; different province

A15 Pailin amphibolite hand specimen | Gap

No directly downloaded locality-verified photograph

A16 Pailin or northeast gneiss hand specimen | Gap

No directly downloaded locality-verified photograph

A17 Schist or low-grade metasediment hand specimen | Gap

JICA basic hornfels or green schist sample listed; no verified matching specimen photo

A18 Pailin crystalline complex outcrop | Gap

No directly downloaded locality-verified photograph

A19 Northeast basement outcrop | Gap

Snoul intrusive outcrop does not establish old crystalline basement

Complete machine-readable record: data/photo_coverage_19_requested_slots.csv. No synthetic photos or locality substitutions.

Sources and reuse conditions

JICA

The master plan study for promotion of the mining industry in the Kingdom of Cambodia, final report, September 2010. Official catalogue and PDFs are retained. Analytical facts were extracted with original units and limits. An open image-reuse license was not verified; field plates are attributed research excerpts. Check permission before public republication.

CCOP and Cambodia MME

Five catalogue records supply seven authentic photographs. The catalogue states CC BY-NC and copyright CCOP. Retain attribution and observe the noncommercial condition. Original metadata HTML is included. Do not assume these photographs may be used commercially.

Peer reviewed and classification sources

Shellnutt et al. 2023, Frontiers in Earth Science, doi:10.3389/feart.2023.1110568, CC BY. Figures 3 and 4 are preserved with panel labels. Gillespie and Styles 1999, BGS RR/99/006, supplies classification references; Crown copyright applies. Do not apply one source license to the entire bundle.

Sources located but not extracted

Uchida et al. 2023 southwestern granite paper: PMC CAPTCHA and publisher 403. Kasahara et al. 2021 author-hosted PDF: 403. Minerals 2018 Ratanakiri zircon paper: 429 rate limit. GEUS/CCOP report 2007/49: verified online, public PDF download timed out twice; no local photographs or chemistry were extracted from it. No access controls were bypassed.

Exact URLs, file sizes, UTC download times and SHA256 checksums: source_downloads.json

Source links and files for checking

JICA catalogue

https://openjicareport.jica.go.jp/660/660/660_109_12006003.html

JICA analytical appendices

https://openjicareport.jica.go.jp/pdf/12006003_05.pdf

CCOP carbonate specimen

https://ccop-gdr.org/data/650cfc4b-d2d2-4c02-8403-5374aefbda55

CCOP sandstone outcrop

https://ccop-gdr.org/data/007fcf1b-615f-4558-b103-21289d3cf168

CCOP weathered basalt

https://ccop-gdr.org/data/723550a9-47f2-4758-aa5f-f79a1dd7b6bd

CCOP lateritic soil

https://ccop-gdr.org/data/e666be92-7ec2-4c43-9ad0-0c23a8ec46b7

Snoul article

https://doi.org/10.3389/feart.2023.1110568

BGS classification source

https://nora.nerc.ac.uk/id/eprint/3223/1/RR99006.pdf

GEUS field report

https://data.geus.dk/pure-pdf/27087_GEUS-R_2007_49_opt.pdf

Source package includes original documents, photographic assets, data CSVs, download receipts and this atlas.